Foundations of Foundation model AI

[Foundation model AI]

1.0 Content Block

Examples TechCrunch saw Sora as an example of a world model, while in January 2025, Nvidia released its own set of world models. The South China Morning Post wrote that Manycore Tech was another example of companies aiming to build a world model, viewing their work as an example of spatial intelligence. In May 2025, Mohamed bin Zayed University of Artificial Intelligence released a world model for building simulations to test AI agents. Google DeepMind has also released two world models in two-dimensional space and three-dimensional space, respectively, that were trained on video data, with Google claiming that the latter can be a training environment for AI agents. Meta released a world model in June 2025, Tencent released an open source world model in July 2025. Niantic, Inc. spinoff, Niantic Spatial, is developing a world model using anonymized player scans from Pokémon GO. Other companies that are planning as of 2025 to build world models include ByteDance and xAI.

2.0 Content Block

History Quanta Magazine traced world models back to a 1943 publication by Kenneth Craik on mental models and the blocks world of SHRDLU in the 1960s. Business Insider traced world models to a 1971 paper by Jay Wright Forrester. A related idea of organizing world knowledge, the frame representation, was proposed by Marvin Minsky in 1974. In 2018, researchers David Ha and Jürgen Schmidhuber defined world models in the context of reinforcement learning: an agent with a variational autoencoder model V for representing visual observations, a recurrent neural network model M for representing memory, and a linear model C for making decisions. They suggested that agents trained on world models in environments that simulate reality could be applied to real world settings. In 2022, Yann LeCun saw a world model (defined by him as a neural network that acts as a mental model for aspects of the world that are seen as relevant) as part of a larger system of cognitive architecture – other neural networks that are analogous to different regions of the brain. In his view, this framework could lead to commonsense reasoning. LeCun has estimated that world models would be fully functional by the late 2020s to mid 2030s.

3.0 Content Block

Modeling For a foundation model to effectively generalize, it must acquire rich representations of the training data. As a result, expressive model architectures that efficiently process large-scale data are often preferred in building foundation models. Currently, the Transformer architecture is the de facto choice for building foundation models across a range of modalities.

4.0 Content Block

General-purpose AI Due to their adaptability to a wide range of use-cases, foundation models are sometimes considered to be examples of general-purpose AI. In designing the EU AI Act, the European Parliament has stated that a new wave of general-purpose AI technologies shapes the overall AI ecosystem. The fuller structure of the ecosystem, in addition to the properties of specific general-purpose AI systems, influences the design of AI policy and research. General-purpose AI systems also often appear in people's everyday lives through applications and tools like ChatGPT or DALL-E. Government agencies like EU Parliament have identified regulation of general-purpose AI, such as foundation models, to be a high priority. General-purpose AI systems are often characterized by large size, opacity, and potential for emergence, all of which can create unintended harms. Such systems also heavily influence downstream applications, which further exacerbates the need for regulation. In regards to prominent legislation, a number of stakeholders have pushed for the EU AI Act to include restrictions on general-purpose AI systems, all of which would also apply to foundation models.

5.0 Content Block

Training Foundation models are built by optimizing a training objective(s), which is a mathematical function that determines how model parameters are updated based on model predictions on training data. Language models are often trained with a next-tokens prediction objective, which refers to the extent at which the model is able to predict the next token in a sequence. Image models are commonly trained with contrastive learning or diffusion training objectives. For contrastive learning, images are randomly augmented before being evaluated on the resulting similarity of the model's representations. For diffusion models, images are noised and the model learns to gradually de-noise via the objective. Multimodal training objectives also exist, with some separating images and text during training, while others examine them concurrently. In general, the training objectives for foundation models promote the learning of broadly useful representations of data. With the rise of foundation models and the larger datasets that power them, a training objective must be able to parse through internet-scale data for meaningful data points. Additionally, since foundation models are designed to solve a general range of tasks, training objectives ought to be domain complete, or able to solve a broad set of downstream capabilities within the given domain. Lastly, foundation model training objectives should seek to scale well and be computationally efficient. With model size and compute power both being relevant constraints, a training objective must be able to overcome such bottlenecks.

6.0 Content Block

World models are sometimes described as foundation models. World models are a representation of an environment intended to predict the state of that environment after taking a set of actions, as well as to implicitly model physical concepts such as gravity. Input prompts for world models can include text or images, as well as videos or 3D scenes, and the resulting 3D environments can be exported. World models, alongside embodied AI, multi-agent models, and neuroscience models of the brain, are seen as alternatives to large language models for achieving general artificial intelligence. World models do not have a fully agreed definition, but have been divided into two scopes: one for representing and understanding the current environment, and another for predicting the future state of that environment. In the former view, world models are developed using model-based reinforcement learning and a Markov decision process, using model predictive control or Monte Carlo tree search to create policies. With the latter, (multimodal) large language models or video generation models can be used. In addition, these environments can be immersive simulations for training AI agents that can interact in the real world.